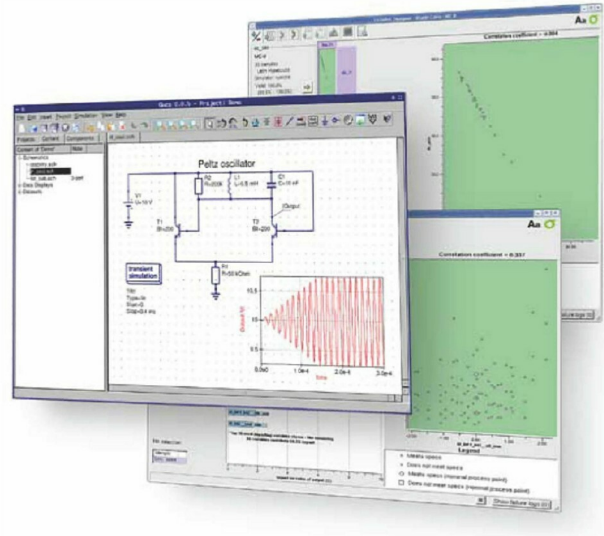
Getting Started with an Open Source Circuit Simulator

Life can be a bit tough without all the electronic gadgets we've got used to. But making an electronic circuit 'work' is even tougher. For electronics hobbyists, there is a lot that they must know before getting their hands dirty, such as circuit designing and debugging. If the design is wrong, most electronic components get damaged, resulting in a waste of money. Featured here is a circuit simulator that hobbyists can get started with.



The Quite Universal Circuit Simulator (QUCS) is an open source electronics circuit simulator software released under the GPL, and is still under development. Though it is very stable, there are things missing here and there, so let's look at what it can and cannot do. The QUCS GUI is based on Qt by Trolltech. The software aims to support all kinds of circuit simulation types, e.g., DC, AC, S-parameter, harmonic balance analysis, noise analysis, etc. The outputs of these analyses can be expressed in various forms like the Smith-Chart, Cartesian, Tabular, Polar, Smith-Polar combination, 3D-Cartesian, Locus Curve, Timing Diagram and Truth Table.

Installing QUCS

On a Debian-based system, QUCS installation is as simple as *sudo apt-get install qucs*, while in Ubuntu QUCS is available in the Ubuntu Software Centre. Or you can do it the hard way, compiling from source. Download the tarball from *http://qucs.sourceforge.net/download.html*